UGANDA CHRISTIAN UNIVERSITY

Master of Science in Data Science and Analytics

Computer Science Seminars and Practicum I-(MSCS-1, MSDS-1)

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Research Topic: Detecting and Mitigating Model Drift in Deployed Healthcare Machine Learning Systems: A Case Study of Sepsis Prediction Drift in ICU Data.

Question:

Literature Review and Problem Formulation

Tasks 1: Research Problem Description

Machine learning models in healthcare are typically trained and validated on retrospective datasets and then deployed into clinical environments that are inherently dynamic. After deployment, these systems operate in settings where patient populations, disease prevalence, clinical guidelines, documentation practices, and workflow structures evolve over time. As a result, the statistical relationship between model inputs and outcomes can shift, even when a model initially demonstrates strong performance. This phenomenon—commonly described as dataset shift or concept drift can lead to gradual and often unnoticed degradation in model reliability (Quiñonero-Candela et al., 2008). The implications are particularly serious in healthcare because deployed models often support high-stakes decisions such as triage prioritization, diagnostic support, and risk stratification. When performance deteriorates without detection, clinical risk increases, clinician trust in decision support may erode, and disparities can widen across hospitals or patient subgroups. In practice, continuous monitoring is further complicated by the limited availability of timely ground-truth labels, since outcomes may be delayed, noisy, or inconsistently recorded. This creates a paradox: the settings where reliability matters most are often those where routine, label-driven validation is operationally constrained (Finlayson et al., 2021). Consequently, detecting and mitigating model drift is not simply a technical optimization problem, but a core requirement for maintaining the safety, equity, and long-term clinical value of deployed machine learning systems. To ground this problem in a clinically important deployment scenario, this study uses **sepsis risk prediction in ICU settings** as a case study. Sepsis models are especially vulnerable to drift because ICU populations and acuity can vary over time, clinical protocols and treatment pathways evolve, and measurement frequency or documentation patterns may change each of which can alter both input distributions and the relationship between predictors and outcomes. This makes sepsis prediction in the ICU a practical and high-impact context for examining how drift can be detected and mitigated under realistic clinical constraints.

Tasks 2: Foundational Literature (Problem Justification)

The theoretical foundations of dataset shift highlight that most predictive models implicitly assume stability between the data distribution observed during training and the distribution encountered after deployment. Quiñonero-Candela et al. (2008) formalize this challenge as a mismatch between training and test joint distributions, explaining why performance estimated

from retrospective evaluation can break down when real-world conditions change. This framework is directly relevant to clinical ML because deployment environments are rarely static for extended periods.

Complementing this perspective, concept drift research emphasizes that the relationship between predictors and outcomes may evolve over time in non-stationary systems (Gama et al., 2014). In such settings, a model that remains unchanged after deployment can become progressively misaligned with the underlying data-generating process. This literature motivates the idea that monitoring and adaptation should be treated as core components of model lifecycle management rather than as optional, post-deployment fixes.

Healthcare-specific evidence reinforces these theoretical concerns and demonstrates that drift-related failures occur in realistic clinical applications. Zech et al. (2018), for example, show that a deep learning model for pneumonia detection can perform strongly in an internal setting yet generalize poorly across hospitals, suggesting that models may inadvertently learn site-specific confounders rather than stable clinical signals. This finding is important for sepsis prediction in ICUs as well, since differences in hospital protocols, lab ordering patterns, documentation workflows, and patient mix can similarly introduce non-clinical signals and shift the model's learned associations over time.

Finally, early clinical deployment discussions emphasize that healthcare ML is exposed to multiple interacting sources of shift such as changes in coding and documentation practices, shifting patient demographics, evolving clinical guidelines, and workflow modifications (Finlayson et al., 2021). However, while foundational work establishes that drift is theoretically expected and empirically observed, it provides limited operational guidance on how health systems should implement continuous monitoring and govern safe adaptation in routine practice.

Collectively, this foundational literature justifies treating drift as an inherent property of deployed healthcare machine learning systems and motivates the need for structured monitoring and controlled updating, particularly for high-impact use cases such as ICU sepsis prediction.

Tasks 3:Current and Recent Research (Problem Currency)

Recent research indicates that dataset shift in healthcare machine learning remains an active and largely operational problem rather than a solved methodological detail. Contemporary clinical AI perspectives increasingly describe drift as an expected feature of real-world deployment, reflecting the reality that health systems continuously evolve in patient mix, clinical practice, and data capture processes (Finlayson et al., 2021). For ICU use cases such as sepsis prediction, this framing is especially relevant because acuity patterns, treatment protocols, and measurement practices can change over relatively short time horizons, creating persistent opportunities for drift.

A consistent theme across recent literature is the gap between proposed mitigation techniques and strong real-world evidence about what works reliably. Systematic reviews focused on temporal dataset shift in clinical medicine identify a broad range of adaptation strategies including periodic retraining, recalibration, and domain adaptation but also highlight limited benchmarking consistency and insufficient comparative evaluation across diverse deployment settings (Guo et al., 2021). This suggests that while the toolbox of drift-mitigation methods is expanding, practical guidance on selecting and validating approaches under heterogeneous clinical conditions remains underdeveloped.

Prospective validation studies further reinforce this concern by documenting performance differences between retrospective evaluation and real-time use. Work examining "performance gaps" during prospective validation shows that deployment conditions often diverge meaningfully from development assumptions, producing measurable degradation that may not be visible during offline testing (Ötleş et al., 2021). These findings matter for sepsis prediction systems because the clinical environment in which sepsis is recognized and treated is sensitive to workflow, staffing, and protocol changes factors that can shift both inputs and outcomes.

Methodological and review research has also clarified that drift can emerge even without overt institutional changes. Sahiner et al. (2023) emphasize subtle mechanisms such as gradual shifts in measurement frequency or evolving clinical decision patterns that can alter the data-generating process and motivate continuous monitoring as a lifecycle requirement. In parallel, empirical work in medical imaging highlights technical limitations of common monitoring practices: Kore et al. (2024) show that predictive performance alone is not a

reliable proxy for drift detection and that detection sensitivity depends on sample size and patient feature composition. This general insight extends to ICU models, where limited sample sizes within short monitoring windows and changing patient composition can complicate drift alarms.

Encouragingly, more recent studies demonstrate the feasibility of proactive monitoring pipelines that can detect and mitigate harmful shifts without depending exclusively on immediate ground-truth labels, supporting label-agnostic or weakly supervised strategies that are better aligned with real hospital constraints (Subasri et al., 2025). However, broader syntheses still report gaps in standardization, evaluation consistency, and operational integration, suggesting that deployment-grade practices remain uneven across the field (Silva et al., 2025).

Overall, the state of the art has progressed from documenting the existence of drift to proposing monitoring architectures and adaptation strategies, yet consistent, evidence-backed deployment patterns particularly in label-scarce settings such as ICU sepsis prediction remain limited, and prospective validation of safe updating practices is still insufficient.

Tasks 4: Gap Analysis (What Is Still Missing)

Despite increasing recognition of dataset shift in healthcare machine learning, several persistent limitations remain, particularly when considering high-acuity use cases such as ICU sepsis prediction. First, many monitoring approaches depend heavily on timely access to reliable ground-truth labels. In ICU settings, however, outcomes such as confirmed sepsis, organ failure progression, or mortality may be delayed, reclassified, or inconsistently documented. This makes continuous, label-driven performance monitoring difficult at deployment scale. As a result, frameworks that rely primarily on real-time outcome feedback risk creating a monitoring blind spot precisely where oversight is most critical (Kore et al., 2024). For sepsis prediction models that operate in time-sensitive environments, this limitation is especially consequential.

Second, drift detection methods often demonstrate instability across contexts. Empirical evaluations show that detection sensitivity depends strongly on sample size, feature distribution, and patient composition (Kore et al., 2024). In ICU settings, where case mix and acuity may fluctuate seasonally or institutionally, a drift detection strategy validated in one

hospital may not perform reliably in another. This variability challenges the assumption that a single detection technique can be universally applied across deployment environments.

Third, prospective validation remains limited. Although retrospective analyses of drift and mitigation strategies are common, relatively few studies evaluate monitoring and adaptation under sustained real-world deployment conditions where workflows, coding practices, and treatment protocols evolve over time (Ötleş et al., 2021). For sepsis models, whose predictions may influence care pathways and resource allocation, the durability of mitigation strategies under changing operational pressures remains insufficiently studied.

Additionally, while retraining, recalibration, and continual learning are frequently proposed as corrective mechanisms, standards for "safe updating" remain under-specified. Questions regarding when to initiate updates, how to validate modified models before redeployment, and how to prevent unintended degradation or subgroup disparities are often discussed conceptually rather than operationally (Finlayson et al., 2021). In high-risk environments such as ICUs, ungoverned updating may introduce new safety and fairness concerns rather than resolving existing drift.

Finally, generalizability across hospitals and patient subpopulations continues to present challenges. Cross-site evidence suggests that models may inadvertently learn confounded or site-specific signals, resulting in external validity failures when deployed in new contexts (Zech et al., 2018). In the case of sepsis prediction, variations in documentation intensity, laboratory ordering patterns, and treatment timing may further amplify these risks, raising both robustness and equity concerns.

Taken together, these limitations highlight a central unresolved need: the development of a practical, label-efficient, and safety-governed framework for drift monitoring and mitigation that can operate reliably across institutions and over time. Such a framework must integrate detection mechanisms, escalation policies, and controlled update validation into a coherent lifecycle strategy, particularly for high-impact clinical applications like ICU sepsis prediction.

Task 5: Proposed Solution

A Label-Efficient Drift Monitoring and Safe-Update Framework for ICU Sepsis Prediction

To address the identified gaps, I propose a deployment-oriented framework that integrates multi-signal drift monitoring with structured, safety-governed model updating, illustrated through the case of ICU sepsis prediction. Rather than introducing a new predictive algorithm, the contribution lies in designing a lifecycle architecture tailored to healthcare's operational constraints particularly label scarcity, high clinical risk, and cross-site variability.

(a) Framework Design

1. Multi-Signal Drift Monitoring (Label-Efficient Layer)

Instead of relying exclusively on outcome-based performance metrics which may be delayed or unreliable in ICU contexts the framework monitors multiple complementary signals:

- **Input distribution monitoring:** Statistical tests such as Kolmogorov–Smirnov tests or Population Stability Index (PSI) can be applied to detect shifts in laboratory values, vital sign distributions, or feature frequency patterns associated with sepsis prediction. These tests target covariate shift before outcome degradation becomes evident.
- **Prediction distribution monitoring:** Monitoring changes in predicted sepsis risk score distributions over time can reveal abnormal inflation, compression, or instability in model outputs that may indicate drift in learned relationships.
- **Uncertainty monitoring:** Confidence measures, entropy-based metrics, or conformal prediction flags can be tracked to detect growing predictive instability, particularly in borderline or high-risk ICU cases.
- **Subgroup monitoring:** Drift signals are evaluated across clinically relevant strata (e.g., age groups, comorbidity burden, hospital site, or admission type) to detect localized degradation that may not appear in aggregate metrics.

By combining these signals, the framework reduces reliance on immediate ground-truth labels and acknowledges that predictive performance alone is often insufficient for reliable drift detection (Kore et al., 2024). Drift alerts require corroboration across multiple indicators, minimizing false alarms and unnecessary retraining.

2. Tiered Action Policy

When monitoring signals exceed predefined thresholds, the framework enforces a structured escalation pathway:

- **Observe:** Intensified monitoring without immediate intervention.
- **Investigate:** Root-cause analysis of affected features, workflows, or subgroups (e.g., changes in sepsis documentation practices or lab ordering frequency).
- **Shadow evaluation:** Candidate recalibrated or retrained models are evaluated in parallel without influencing live clinical decisions.
- **Update:** Validated models are promoted to active deployment.

This tiered structure prevents automatic retraining in response to minor fluctuations and reduces the risk of destabilizing a clinically integrated sepsis model.

3. Safe Updating Mechanism (Governed Adaptation Layer)

Updates follow a conservative hierarchy designed for clinical safety:

- **Recalibration first**, as it adjusts probability estimates without altering learned feature relationships, reducing structural risk.
- **Constrained retraining** only when recalibration fails to restore stability.

Before promotion, updated models must satisfy predefined acceptance criteria, including:

- Maintenance of overall discrimination and calibration
- Stability across key ICU subgroups
- Robustness under simulated shift scenarios
- No evidence of new subgroup disparities

Version control, audit logging, and rollback protocols are mandatory to ensure reversibility if post-update degradation is observed.

(b) How the Framework Addresses Identified Gaps

This framework directly addresses the limitations identified in recent literature. By prioritizing label-efficient monitoring signals, it mitigates dependence on real-time outcome labels a central constraint in ICU settings (Kore et al., 2024; Subasri et al., 2025). Multi-signal corroboration responds to evidence that performance metrics alone are insufficient for drift detection (Kore et al., 2024). Furthermore, the explicit tiered escalation and acceptance testing structure operationalizes "safe updating" as a governed lifecycle process rather than an ad hoc technical adjustment, aligning with calls for structured clinical AI oversight (Finlayson et al., 2021).

(c) Novelty and Expected Contribution

The novelty of this proposal lies not in introducing a new drift detection algorithm, but in integrating empirically grounded detection methods with governance-driven update controls into a coherent deployment pipeline for ICU sepsis prediction. It responds directly to three consistent findings in the literature:

1. Drift in healthcare ML is common and inevitable.
2. Detection without timely labels is technically challenging.
3. Safe updating lacks standardized operational structure.

By explicitly linking drift signals to tiered mitigation actions and predefined validation thresholds, the framework offers a practical, clinically realistic lifecycle design. In doing so, it aims to bridge the gap between methodological research on dataset shift and the operational demands of sustaining reliable sepsis prediction models in dynamic ICU environments (Finlayson et al., 2021; Silva et al., 2025).

In summary, the literature indicates that model drift is an expected and empirically observed feature of deployed healthcare machine learning, driven by changes in patient populations, clinical practice, and data-generation processes over time. Although foundational and recent research has clarified drift mechanisms and proposed mitigation strategies, persistent gaps remain especially the reliance of monitoring on timely labels, instability of drift detectors across contexts, limited prospective benchmarking, and the absence of clearly operationalized standards for safe updating and governance. Motivated by these gaps, this work proposes a label-efficient, multi-signal drift monitoring and guardrailed update framework, using ICU

sepsis prediction as a clinically meaningful case study. By linking drift evidence to tiered operational responses and explicit acceptance criteria, the framework treats monitoring and updating as a governed lifecycle process rather than a one-time deployment event, with the goal of sustaining reliability, safety, and equity in real clinical environments.

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